Azure Data Scientist Training

DP-100 Certification Prep — Day 1

A hands-on, enterprise-focused training for IT professionals and cloud engineers ready to design and implement machine learning solutions on Azure.

Course Introduction

Training Goals & What You'll Learn

Understand Azure ML

Navigate the Azure Machine Learning workspace, studio, and core components.

Build ML Workflows

Design end-to-end pipelines — from data ingestion through model deployment.

Pass DP-100

Align your skills with the exam objectives for the Azure Data Scientist Associate certification.

Hands-On Labs

Apply concepts immediately with guided labs using real Azure environments.

DP-100 Certification

What the Exam Measures

Design & Prepare ML Solutions

Choose the right Azure services and environments for ML workloads.

Explore & Process Data

Work with datasets, datastores, and feature engineering pipelines.

Train & Optimize Models

Run experiments, tune hyperparameters, and evaluate model performance.

Deploy & Monitor

Register, deploy, and monitor ML models using Azure endpoints.

Day 1 Agenda

Today's Roadmap

01

Data Science Fundamentals

AI, ML, Deep Learning, and real-world use cases

02

Statistics & ML Overview

Key stats concepts, supervised vs. unsupervised learning, algorithms

03

Data Science Lifecycle

From data collection to deployment and monitoring

04

Azure ML Platform

Architecture, studio UI, SDK, CLI, and ecosystem comparison

05

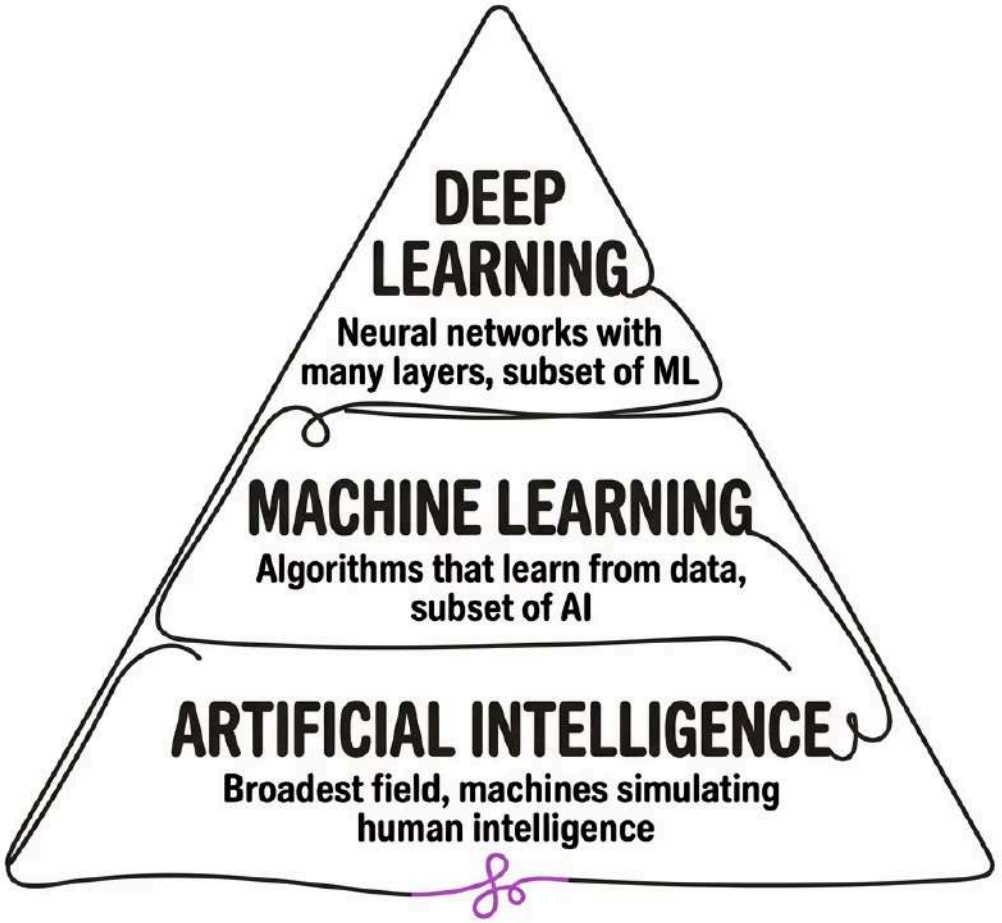
Hands-On Labs 1–3

Workspace setup, compute resources, and notebook environment

What Is Data Science?

Data Science is the discipline of extracting actionable insights from raw data using statistics, programming, and domain expertise. It sits at the intersection of mathematics, software engineering, and business strategy — enabling organizations to make smarter, faster decisions at scale.

AI, Machine Learning & Deep Learning



Artificial Intelligence

The broad field of building machines that mimic human reasoning, perception, and decision-making.

Machine Learning

A subset of AI where models learn patterns from data rather than following explicit rules.

Deep Learning

A subset of ML using multi-layered neural networks — ideal for images, audio, and text.

Data Science in Practice

Real-World Use Cases

Fraud Detection

Banks use ML to flag anomalous transactions in real time, reducing fraud losses by up to 40%.

Recommendation Systems

Streaming and e-commerce platforms drive engagement with personalized content powered by collaborative filtering.

Churn Prediction

Telecom and SaaS companies predict which customers will cancel, enabling proactive retention campaigns.

Statistics for ML

Why Statistics Matters

> Without statistics, machine learning models are black boxes. Understanding the math builds trust in your results and helps you diagnose failures.

Mean & Std Dev

Measure central tendency and spread — key for feature scaling and normality checks.

Correlation

Identifies linear relationships between features — informs feature selection.

Probability

Underpins classification outputs, Bayesian models, and uncertainty estimation.

Statistics in Action

Key Concepts at a Glance

Mean (μ)

Average value of a feature. Used in normalization. Sensitive to outliers.

Standard Deviation (σ)

Measures data spread. High σ = high variance. Low σ = tightly clustered data.

Correlation (r)

Ranges from -1 to +1. Strong correlations reveal important predictive features.

Probability (P)

Models output probabilities. Thresholds determine class predictions in classification.

Machine Learning Overview

Supervised · Unsupervised · Deep Learning

Machine Learning Overview

Supervised vs. Unsupervised Learning

Supervised Learning

Trains on **labeled data** — the correct answer is provided during training.

- Classification (spam vs. not spam)
- Regression (predicting house prices)
- Requires curated, annotated datasets

Unsupervised Learning

Trains on **unlabeled data** — the model discovers hidden patterns.

- Clustering (customer segments)
- Dimensionality reduction (PCA)
- Anomaly detection

Machine Learning Overview

Classification vs. Regression

Classification	Regression
Output: A discrete category or class label	**Output:** A continuous numeric value
Example: Is this email spam? Will this customer churn?	**Example:** Predict sales revenue, forecast temperature
Metric: Accuracy, Precision, Recall, F1	**Metric:** MAE, RMSE, R^2 Score

Machine Learning Overview

Common ML Algorithms

Linear & Logistic Regression

Baseline algorithms for regression and binary classification. Fast, interpretable, great starting points.

Decision Trees

Rule-based splits on features. Highly interpretable but prone to overfitting on complex data.

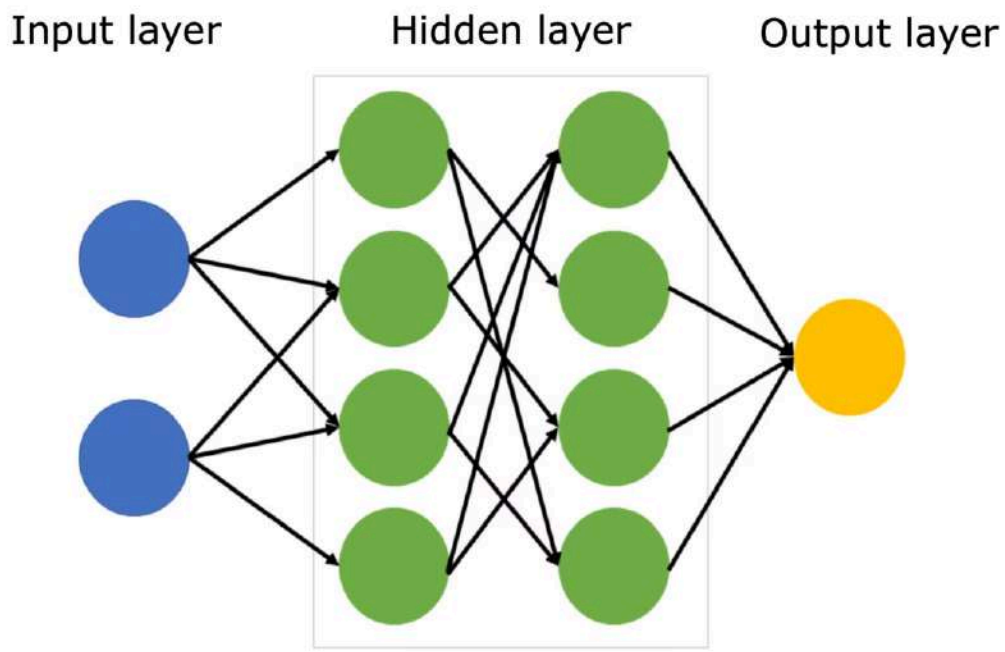
Random Forest

Ensemble of decision trees using bagging. Reduces variance and outperforms single trees.

Gradient Boosting

Sequential ensemble method (XGBoost, LightGBM). State-of-the-art for tabular data tasks.

Deep Learning — When to Use It



Artificial neural networks

Deep learning excels when data is **unstructured and abundant**. It requires significant compute and data volume to outperform traditional ML.

Computer Vision

Image classification, object detection, medical imaging

Natural Language Processing

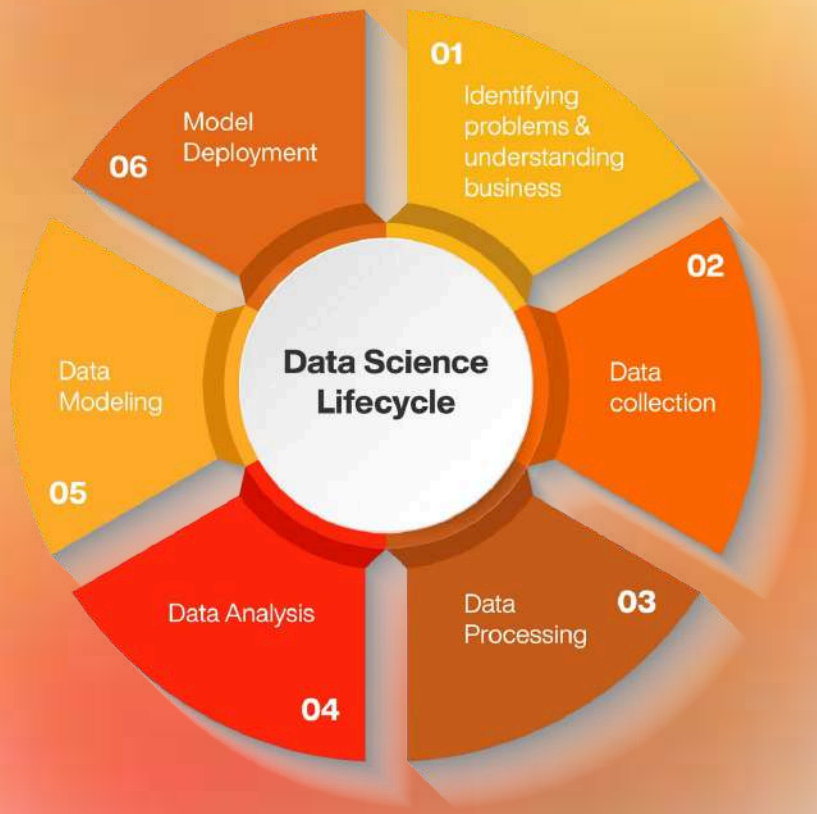
Sentiment analysis, translation, chatbots (LLMs)

Time-Series & Speech

Demand forecasting, voice recognition, anomaly detection

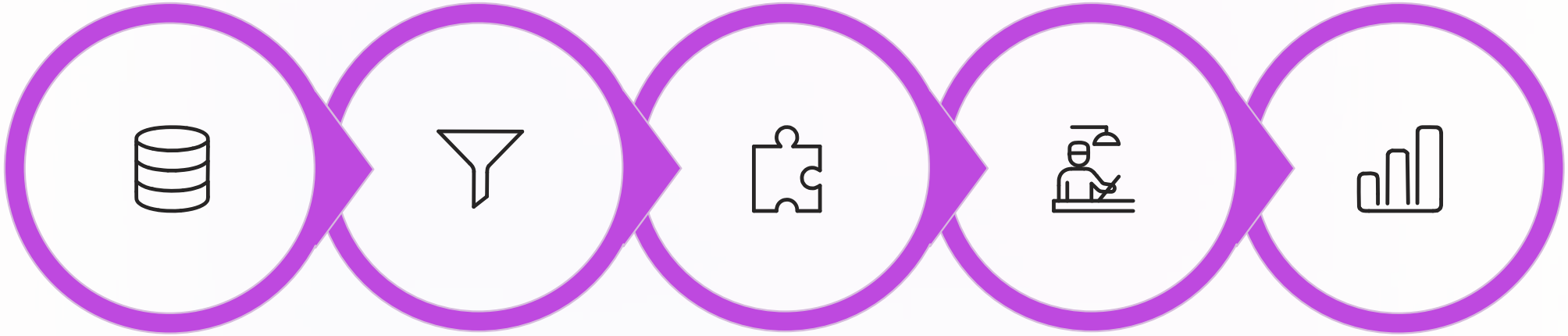
The Data Science Lifecycle

From Raw Data to Production-Ready Models



Data Science Lifecycle

End-to-End Workflow



Data Collection → **Data Preparation** → **Feature Engineering** → **Model Training** → **Model Evaluation**

Each phase feeds into the next — and monitoring feedback loops back to collection, making this a continuous improvement cycle rather than a one-time process.

Data Science Lifecycle

Phase 1 & 2: Collect & Prepare Data

Data Collection

- Identify internal databases, APIs, logs
- Define labeling strategy for supervised tasks
- Assess data volume and quality

Data Preparation

- Handle missing values (imputation or removal)
- Encode categorical variables
- Remove duplicates, outliers, corrupted records

Rule of thumb: Data scientists spend 60–80% of project time on data collection and preparation. Clean data is the most valuable asset in any ML project.

Data Science Lifecycle

Phase 3 & 4: Feature Engineering & Training

Feature Engineering

Transform raw data into features that improve model performance.

- Create ratio features (e.g., spend per visit)
- Bin continuous variables into buckets
- Normalize/standardize numeric columns
- Extract date parts (day of week, month)

Model Training

Split data, select algorithm, and fit the model.

- Train/validation/test split (e.g., 70/15/15)
- Choose baseline algorithm first
- Track experiments with metrics and parameters
- Use cross-validation to reduce bias

Data Science Lifecycle

Phase 5–7: Evaluate, Deploy & Monitor

Model Evaluation

Compare metrics across experiments. Validate on held-out test set. Check for bias and overfitting.

Deployment

Register the model, containerize it, and expose it via a REST endpoint for real-time or batch scoring.

Monitoring

Track prediction drift, data drift, and latency. Trigger retraining pipelines when model degrades.

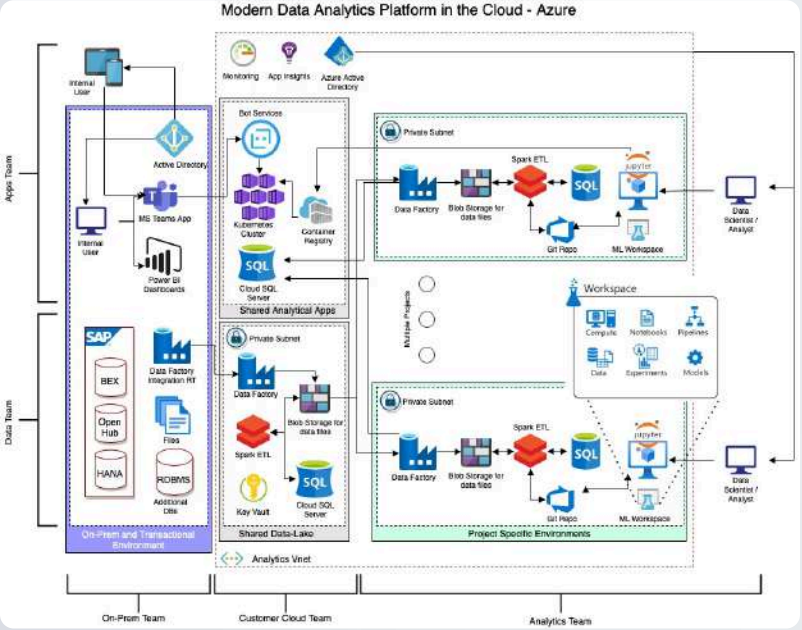
Lifecycle Example

Customer Churn Prediction — End to End

1. **Collect**
   CRM data, usage logs, billing history, support tickets
2. **Prepare**
   Fill missing tenure values, encode contract type
3. **Engineer**
   Days since last login, monthly spend trend, NPS score
4. **Train**
   Gradient Boosting classifier on 80/20 split
5. **Deploy**
   Azure ML real-time endpoint, scored nightly
6. **Monitor**
   Alert when data drift exceeds threshold; retrain monthly

Why Azure Machine Learning?

Solving Real ML Engineering Challenges at Enterprise Scale



Why Azure ML

The Challenges Without a Platform

No Experiment Tracking

Teams lose track of which model, parameters, and data version produced the best result. Reproducibility breaks down.

Infrastructure Overhead

Setting up GPU clusters, managing environments, and scaling training jobs manually consumes weeks of engineering time.

Deployment Complexity

Packaging models, managing containers, and exposing secure endpoints requires DevOps skills most data scientists lack.

No Governance

Without a registry, organizations can't audit model versions, track lineage, or enforce access controls across teams.

Why Azure ML

How Azure ML Solves These Challenges

Experiment Tracking

Automatic logging of runs, metrics, parameters, and artifacts in a central MLflow-compatible registry.

Managed Compute

Spin up compute instances and auto-scaling clusters in minutes. Pay only when running.

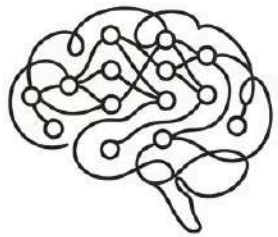
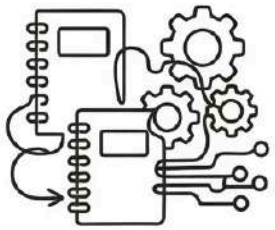
One-Click Deployment

Deploy models to managed online or batch endpoints with built-in container management and scaling.

Model Governance

Version-controlled model registry with lineage tracking, responsible AI dashboards, and RBAC support.

Choosing the Right Azure ML Service

AZURE ML	AZURE DATABRICKS	AZURE SYNAPSE ANALYTICS
		
ML TRAINING	BIG DATA	WAREHOUSING
DATA SCIENTISTS	ENGINEERS	ANALYSTS
AutoML, MLOps	DELTA LAKE, SPARK	SQL POOLS, POWER BI

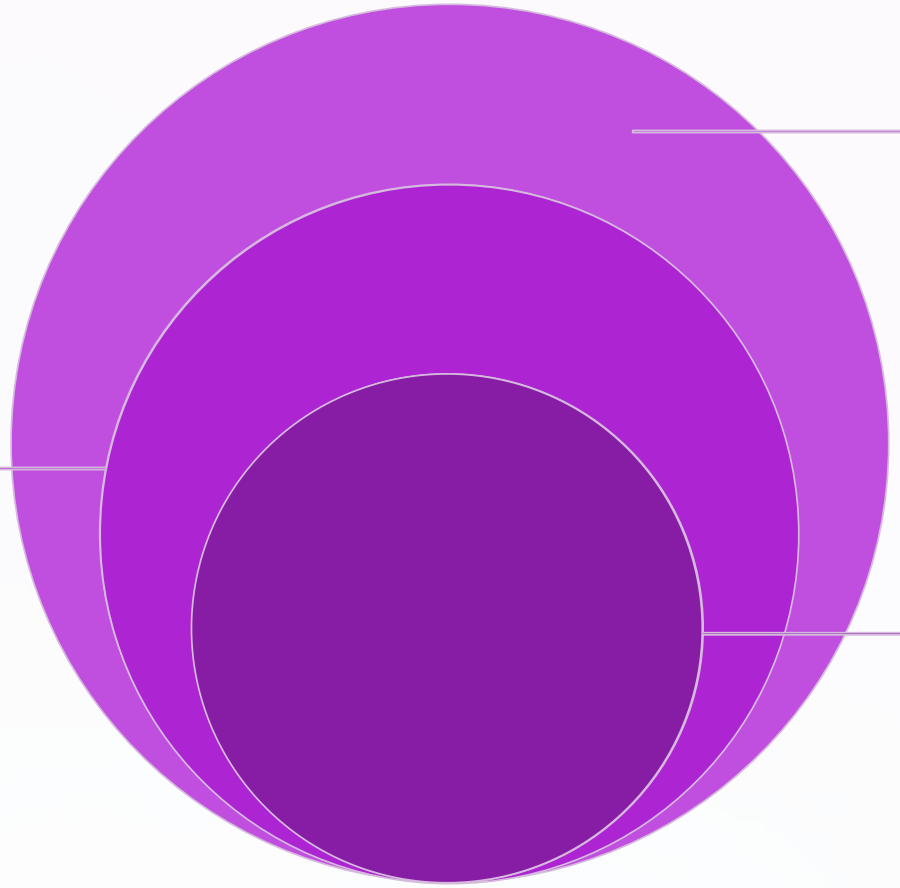
Which service should you use?

- **Azure ML:** Your primary workspace for DP-100 — training, tracking, and deploying models
- **Databricks:** When data volumes require Spark and collaborative data engineering
- **Synapse:** For enterprise SQL analytics and data warehouse integration

These services are **complementary**, not competing. Most enterprise architectures use all three together.

Azure Data Platform Ecosystem

Reference Architecture



Ingest & Store

Data Factory, Event Hubs, and ADLS Gen2

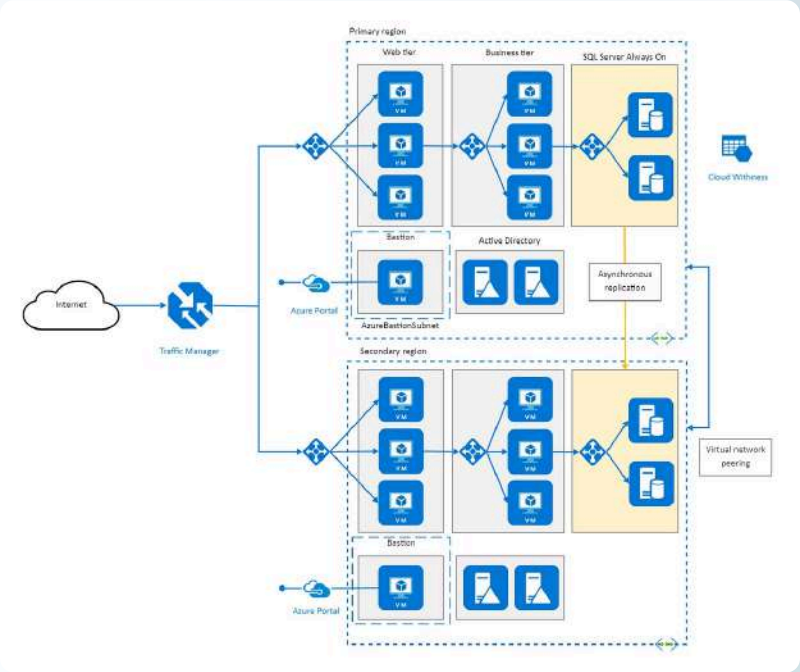
Process

Azure Databricks ETL and Synapse SQL analytics

Model & Serve

Azure ML for training and endpoints

Data flows from ingestion through storage and processing before reaching Azure ML — where models are trained on prepared, clean features and served to production applications via endpoints.



Azure Machine Learning Architecture

Core Components & How They Connect

Azure ML Architecture

Core Workspace Components

Workspace

The top-level resource. Contains all assets: data, compute, models, experiments, and pipelines.

Compute Instance

A managed cloud VM for interactive development. Runs Jupyter notebooks and VS Code in the browser.

Compute Cluster

Auto-scaling pool of nodes for training jobs. Scales to zero when idle — pay only when running.

Datasets & Datastores

Versioned references to data in Azure Storage. Decouples data access from training scripts.

Azure ML Architecture

MLOps Components

Experiments

Logical groupings of training runs. Each run logs metrics, parameters, outputs, and environment details.

Model Registry

Versioned store of trained models with metadata, tags, and lineage tracking for governance.

Endpoints

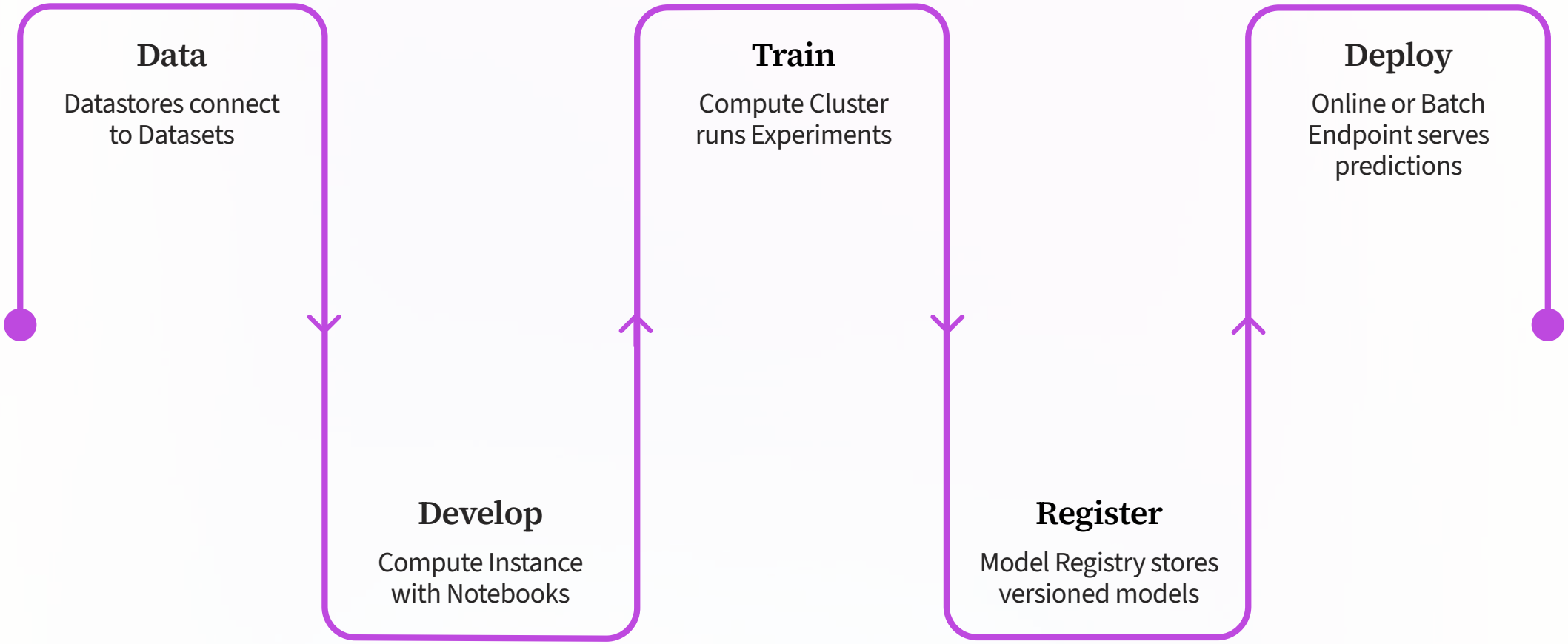
Managed online endpoints for real-time scoring or batch endpoints for large-scale inference jobs.

Pipelines

Reusable, parameterized ML workflows. Each step runs independently and can be cached for efficiency.

Azure ML Architecture

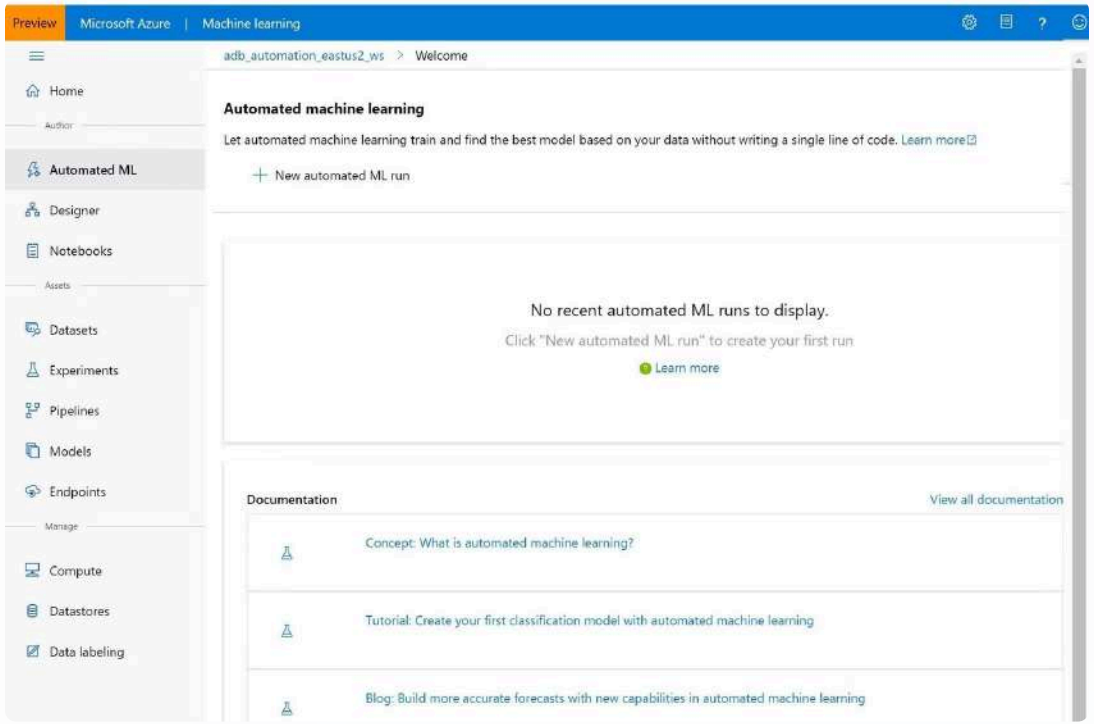
Workspace Visual Workflow



Every Azure ML project follows this flow — from connecting data sources through interactive development, scalable training, model registration, and finally production deployment.

Azure ML Studio

Navigating the Studio UI



- **Home**

  Recent activity, quick links, and workspace overview

- **Data & Datastores**

  Register, version, and explore datasets

- **Compute**

  Manage instances, clusters, and attached compute

- **Experiments & Models**

  Browse run history, compare metrics, register models

- **Endpoints & Notebooks**

  Deploy models and write code interactively

SDK vs CLI vs Studio

Choosing Your Interaction Method

Feature	Studio (UI)	Python SDK v2	Azure CLI (ml ext)
Best for	Exploration & monitoring	Code-first development	CI/CD automation
Learning curve	Low	Medium	Medium
Automation	Limited	Full programmatic control	Pipeline & DevOps ready
Typical user	Data scientists, analysts	ML engineers, data scientists	MLOps engineers, DevOps

For DP-100, you must be proficient in **all three**. Studio for exploration, SDK for training scripts, CLI for deployment automation.

SDK vs CLI vs Studio

Python SDK v2 — Quick Reference

The Azure ML Python SDK v2 is the primary tool for building production ML workflows. Here's how to connect and submit a basic job:

```
from azure.ai.ml import MLClient
from azure.identity import DefaultAzureCredential

# Connect to the workspace
ml_client = MLClient(
 DefaultAzureCredential(),
 subscription_id="<your-sub>",
 resource_group_name="rg-azureml",
 workspace_name="ws-dp100"
)

# List registered models
for model in ml_client.models.list():
 print(model.name, model.version)
```

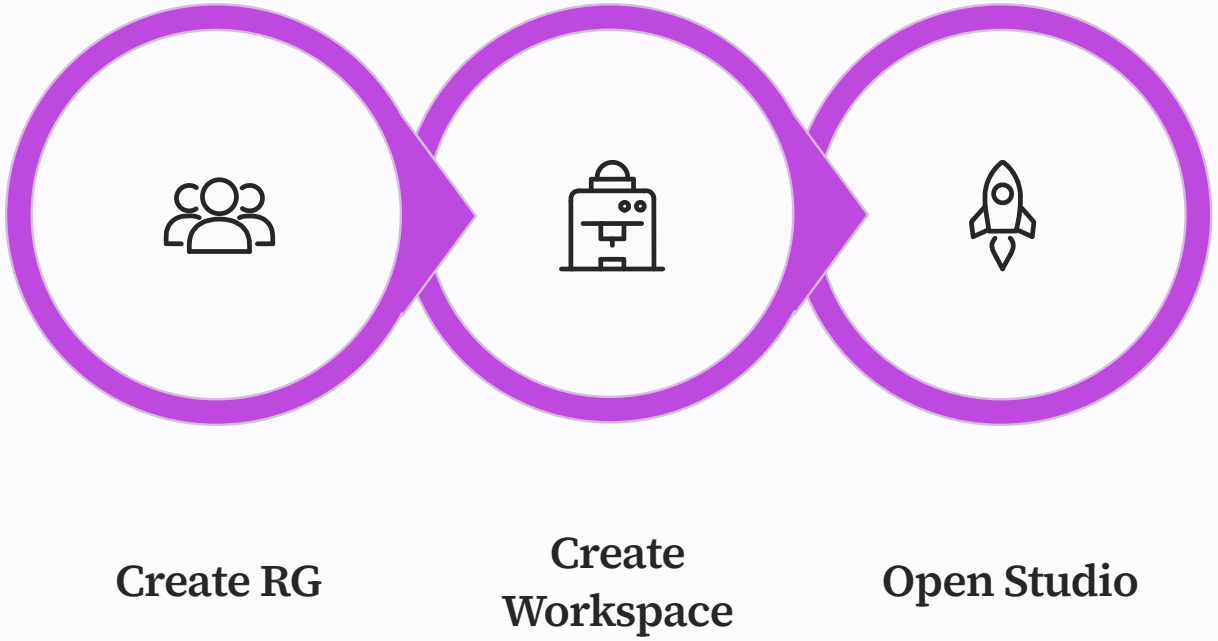
Hands-On Lab 1

Create Your Azure ML Workspace

Estimated time: **20 minutes** — Azure Portal required

Lab 1 — Azure ML Workspace

Step-by-Step Instructions



Once your workspace is created, all subsequent labs run inside it. Use a dedicated resource group to simplify cleanup and cost tracking at the end of the course.

Lab 1 — Azure ML Workspace

What Gets Created Automatically

Azure Storage Account

Stores datasets, model artifacts, run outputs, and experiment logs.

Azure Key Vault

Securely stores secrets, credentials, and connection strings used by the workspace.

Azure Container Registry

Stores custom Docker images for training environments and deployment containers.

Application Insights

Monitors deployed endpoints for request volume, latency, and error rates.

We added a node through Windows Admin Center to demonstrate the ease of scaling out using Dell OpenManage Integration with Windows Admin Center. We initially deployed a 3-node cluster and added a fourth node prior to testing. Before adding a node to the cluster, we needed to make sure the node was joined to the Active Directory domain and added as a connection in Windows Admin Center.

Figure 15 and Figure 16 show how we used this feature for our cluster.

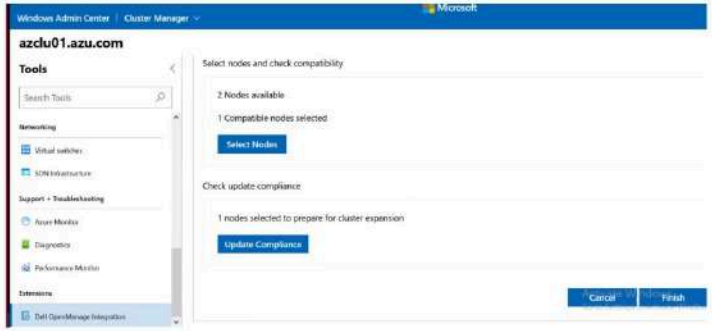


Figure 15. Select node to check compliance

Hands-On Lab 2

Create Compute Resources

Estimated time: **20 minutes**

Lab 2 — Compute Resources

Compute Instance vs. Compute Cluster

Compute Instance

Purpose: Interactive development

- Single-node managed VM
- Always-on or scheduled shutdown
- Runs Jupyter, VS Code, RStudio
- Recommended: **Standard_DS3_v2**

Compute Cluster

Purpose: Scalable training jobs

- Multi-node, auto-scaling pool
- Scales to zero when idle
- Supports distributed training
- Recommended: **Standard_DS3_v2** (CPU), **Standard_NC6** (GPU)

Lab 2 — Compute Resources

Autoscaling & CPU vs. GPU

Autoscaling Explained

Set **min nodes = 0** and **max nodes = 4**. The cluster scales up when a job is submitted and back to zero when idle — eliminating idle compute costs entirely.

When to Use CPU Compute

Traditional ML algorithms (Scikit-learn, XGBoost, LightGBM) run efficiently on CPU. Use CPU clusters for tabular data tasks — they're cheaper and faster to provision.

When to Use GPU Compute

Reserve GPU compute (NC or ND series VMs) for deep learning workloads — image models, transformers, and large neural networks. GPUs can be 10–50× faster for these tasks.

Lab 2 — Compute Resources

Create Compute Cluster via CLI

```
# Create a CPU compute cluster with autoscaling
az ml compute create \
  --name cpu-cluster \
  --type AmlCompute \
  --min-instances 0 \
  --max-instances 4 \
  --size Standard_DS3_v2 \
  --resource-group rg-azureml \
  --workspace-name ws-dp100
```

Cost tip: Always set `min-instances 0` in training clusters. A cluster sitting at 1+ idle nodes will continuously accrue compute charges even when no jobs are running.

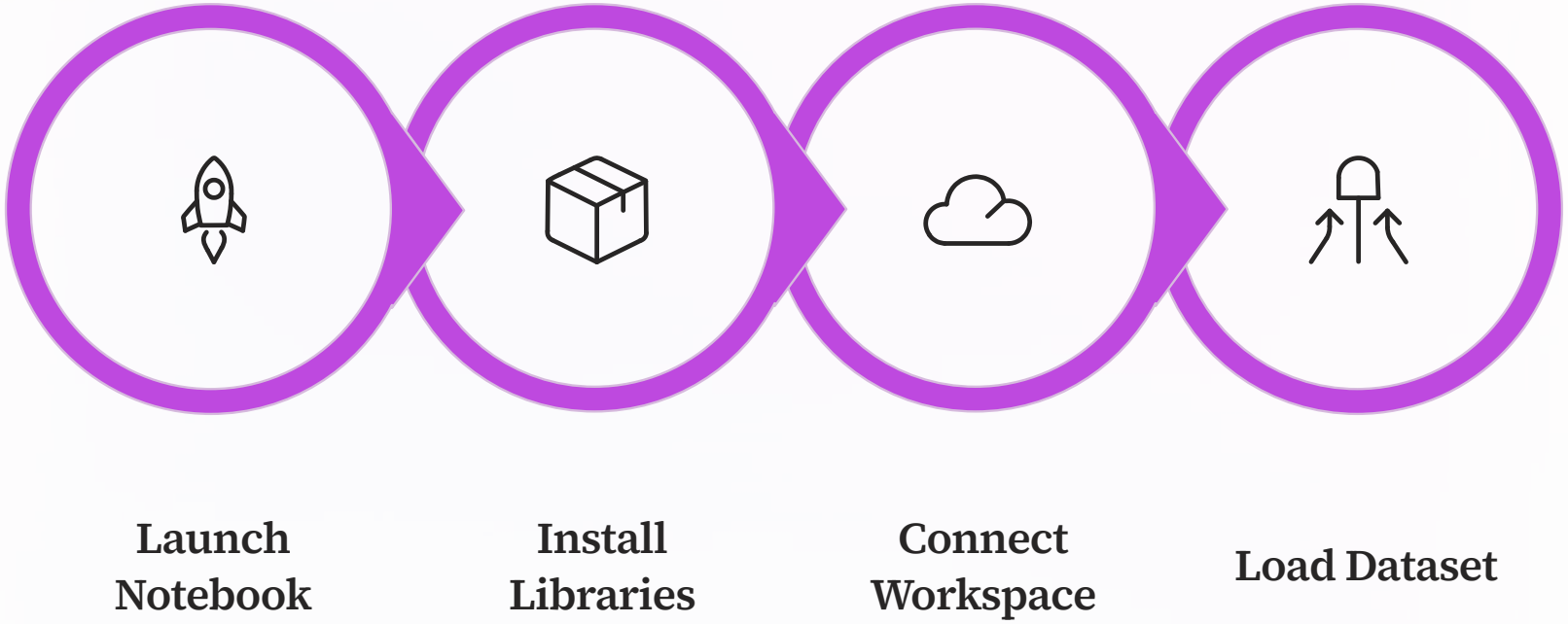
Hands-On Lab 3

Azure ML Notebook Environment

Estimated time: **30 minutes**

Lab 3 — Notebook Environment

Launch & Configure Jupyter Notebooks



The Azure ML notebook environment provides a managed, pre-configured Python environment with persistent storage — no local setup required for development.

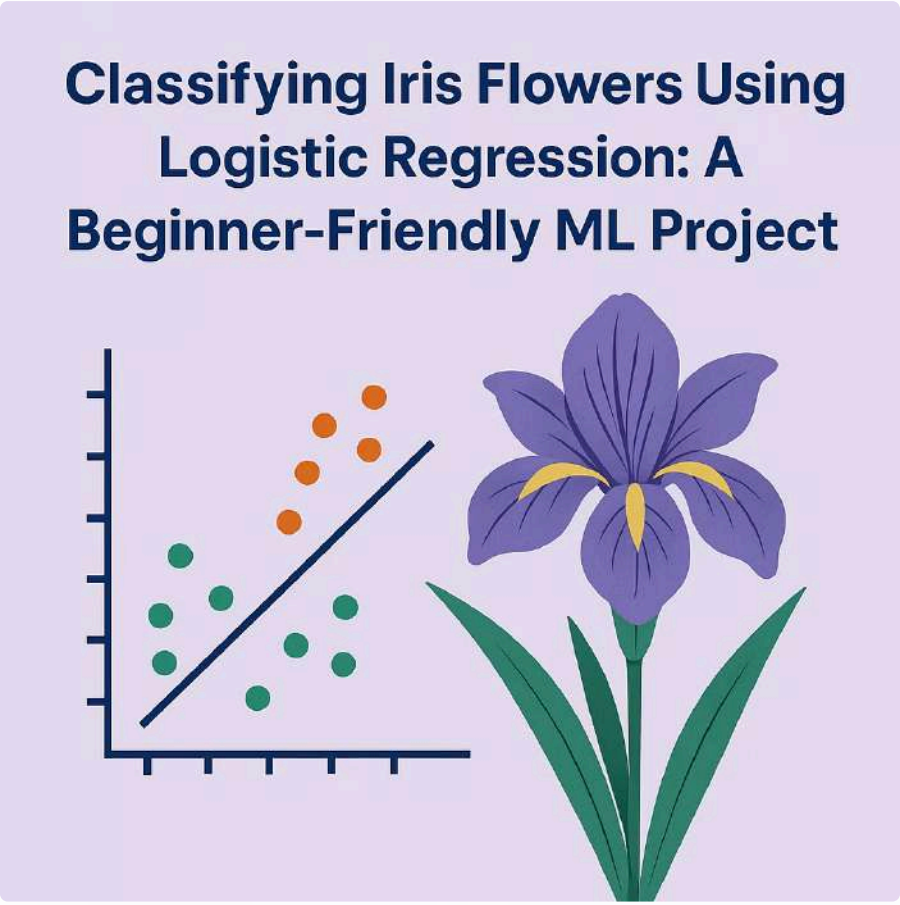
Lab 3 — Notebook Environment

Load & Explore the Iris Dataset

```
from azure.ai.ml import MLClient
from azure.identity import DefaultAzureCredential
import pandas as pd
from sklearn.datasets import load_iris

# Connect to workspace
ml_client = MLClient(
 DefaultAzureCredential(),
 subscription_id="<your-sub>",
 resource_group_name="rg-azureml",
 workspace_name="ws-dp100"
)

# Load Iris dataset
iris = load_iris(as_frame=True)
df = iris.frame

print(df.head())
print(df.describe())
print(df['target'].value_counts())
```

Lab 3 — Notebook Environment

About the Iris Dataset



The Iris dataset is the classic starting point for classification — small, clean, and well-understood.

In today's lab, you'll verify connectivity to your workspace, run basic EDA, and confirm your environment is ready for Day 2 training exercises.

150 samples
50 per class, 3 flower species

4 features
Sepal/petal length and width in cm

3 classes
Setosa · Versicolor · Virginica

Cost Optimization Tips

Keep Your Azure Bills Under Control

Cost Optimization

Best Practices for Training Environments

Stop Compute Instances

A running compute instance costs money 24/7. Set an auto-shutdown schedule (e.g., 7 PM daily) or stop it manually after each session.

Min Nodes = 0

Always configure training clusters with minimum node count of 0. The cluster scales up only when a job is submitted and returns to zero when idle.

Avoid GPU in Dev

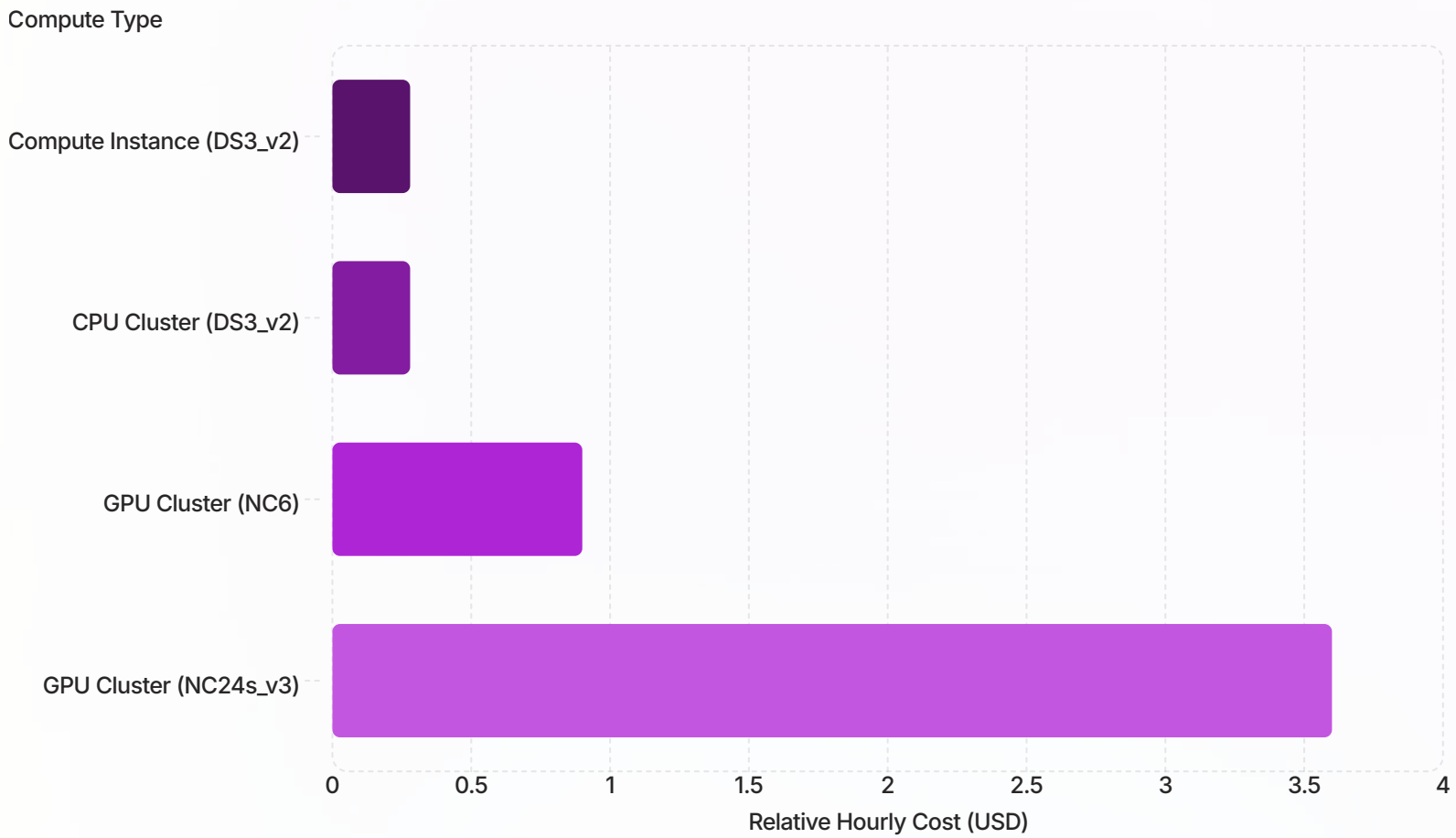
GPU VMs cost 5–10× more than CPU VMs. Use CPU compute instances during development and reserve GPU clusters for final training runs only.

Set Spending Alerts

Configure Azure Cost Management alerts at 80% and 100% of your budget. Receive email notifications before unexpected overages occur.

Cost Optimization

Quick Reference: Cost by Compute Type



Idle GPU clusters and always-on compute instances are the top two sources of unexpected Azure ML cost overruns in training environments.

Day 1 Summary

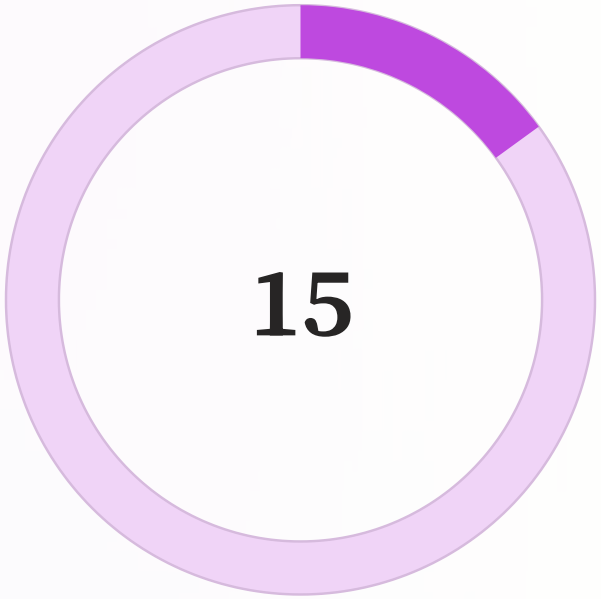
What We Covered Today

Core Sections

Data Science, Statistics, ML Overview, Lifecycle, Azure ML Platform, Labs

Hands-On Labs

Workspace, Compute Resources, Notebook Environment



Key Concepts

From AI/ML fundamentals to Azure architecture and SDK basics

Day 1 Summary

Key Takeaways

1 **Azure ML is the DP-100 platform**

Workspace, compute, experiments, registry, and endpoints form a complete MLOps system.

2 **The lifecycle is iterative**

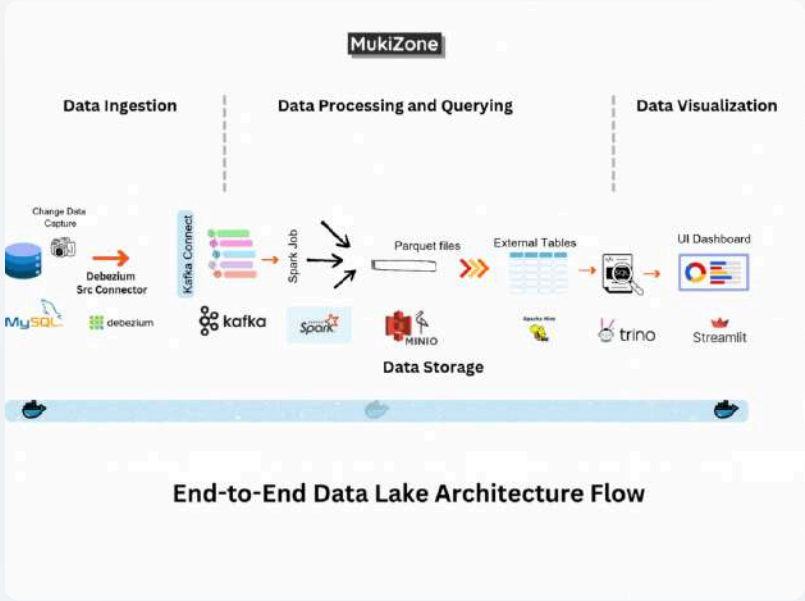
Monitoring feedback loops back to data collection — ML is never truly "done" in production.

3 **SDK + Studio + CLI are complementary**

Use Studio to explore, SDK to build, and CLI to automate. Know when to use each.

4 **Cost discipline starts on Day 1**

Set min nodes to zero, stop instances when done, and reserve GPU for production training.



End-to-End Data Lake Architecture Flow

Coming Up: Day 2

Datasets, Datastores & Data Preprocessing

Preview — Day 2

What's Next

01

Datastores & Datasets

Register Azure Blob/ADLS datastores, create versioned MLTable and URI File datasets, and consume them in training jobs.

02

Data Preprocessing Pipelines

Build reusable feature engineering pipelines with Scikit-learn transformers and Azure ML Pipeline steps.

03

Responsible Data Practices

Handle class imbalance, apply data drift detection, and implement privacy-aware data handling.

04

Lab: Data Prep with Azure ML

Register a real dataset, build a preprocessing pipeline, and output a cleaned training dataset to Azure Storage.